Master Degree Project

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Comprehensive Analysis of lncRNA and circRNA Mediated ceRNA network in Psoriasis

Master's degree Project in bioinformatics
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Abstract

Evidence is accumulating that noncoding RNAs and circRNA are involved in psoriasis; however, the competing endogenous RNA (ceRNA) mediated regulatory mechanisms in psoriasis are rarely reported. The research study aimed to comprehensively investigate the differences in the expression levels of circular RNA (circRNA), long non-coding RNA (lncRNA), microRNA (miRNA/miR), and mRNA in psoriasis. In addition, key lncRNA/circRNA-miRNA-mRNA-ceRNA interactions were screened using the GSE145305 microarray dataset from the Gene Expression Omnibus database. After data preprocessing, differentially expressed circRNAs (DECs), lncRNAs (DELs), miRNAs (DEMs), or genes (DEGs) were identified, and normal controls using the linear models for the microarray data method. A protein-protein interaction (PPI) network was constructed for DEGs based on protein databases, followed by a module analysis. The ceRNA network was constructed based on the interaction between miRNAs and mRNAs and lncRNAs/circRNAs and miRNAs. The present study identified that in the case of mRNA 10 genes are significantly down-regulated, 86 genes are significantly up-regulated and in the case of miRNA 48 are significantly down-regulated and 75 genes are significantly up-regulated between patients with psoriasis and controls. miRNA, mRNA, lncRNA, and circRNA target predictions were made. Then combined construction of a ceRNA network using mRNA-miRNA-lncRNA and mRNA-miRNA-circRNA. The current research has employed the knowledge of bioinformatics tools and software to determine the hub module and PPI network. Taken together, these identified ceRNA interactions may be crucial targets for the treatment of psoriasis.

1-Introduction

Psoriasis is a chronic autoimmune skin disease characterized by scaly erythema and regulated by genetic and environmental agents. Although psoriasis is not life-threatening, it can impose a serious psychological and economic burden on patients. Fortunately, with a better understanding of psoriasis, more and more therapies have emerged. Due to the unclear scientific cause and complex pathogenesis of psoriasis, it remains incurable. Therefore, there is a great need for understanding the molecular networks involved in psoriasis to improve the quality of psoriasis care [1]. Psoriasis is a type of skin disease that creates skin cells that multiply up to 10 times faster than normal cell growth. As a result, bumpy red patches covered with white scales form on the skin and are severely irritated. They can grow anywhere, but most seem to be on the scalp, elbows, knees, and lower back. It is a disease that cannot be transmitted from one person to another. It sometimes occurs in members of an identical family [2,3]. It could have inherited characteristics.

Psoriasis is a chronic inflammatory dermatological disease. It has a potentially strong genetic propensity to be inherited by the next generation. It has autoimmune pathogenic properties. There are the following common types of psoriasis reported [4]. Plaque psoriasis is a very popular form, affecting approximately 90% of people with psoriasis. In plaque psoriasis, lesions are broadly demarcated red plaques covered by silvery scales. Guttate psoriasis is characterized by red droplets that can be widespread. The lesions are often located on the trunk but can later appear anywhere on the body. Presents as blisters of non-infectious pus. Blisters can be localized pustular psoriasis affecting, for example, palms and soles, or generalized with dark red patches [5].

The most common is psoriasis vulgaris, also known as plaque psoriasis. Although it is a dermatological criterion, psoriasis also affects the joints [6]. Some other infections and physiological conditions also influence the pathogenesis of the disease. Hypertension, coronary heart disease, type 2 diabetes, hyperlipidemia, etc. are some significant physiological conditions associated with this disease [7].
The disease has both physical and psychological burdens. The main symptoms include fainting, itching, and bleeding. The disease is seen equally in both sexes [8]. Early detection of the disease is not yet very likely because it starts with a simple rash and therefore cannot be differentiated. The molecular reason for these autoimmune criteria is not yet entirely clear to scientists. Some important details about psoriasis disease are:

- Psoriasis is a chronic immune-modulated inflammatory skin disease with abundant phenotypic subtypes such as guttate and plaque.
- It is extremely hard to cure the psoriasis sites such as the face, nails, and palms.

- Psoriasis has an inherited incidence of 70 to 80%.
- Treatment includes topical therapies, phytotherapy, psoralen, and ultraviolet [9].

MiRNAs are ~22 nucleotides long single-stranded endogenous RNA that does not code for any protein [2]. They can bind to the seed region of mature target mRNAs and adversely regulate the expression of these genes through translational inhibition or degradation of the mRNAs [10]. Reports have confirmed the regulatory role of miRNA in cellular differentiation, maturation, apoptosis, and other signaling pathways [11]. It can also cause dysregulation of the gene expression profile when expressed abnormally [5]. Although a wide study on the pathogenesis of psoriasis has been conducted; the mechanisms of psoriasis have yet to be fully elucidated. Studies show that lncRNA plays a crucial role in psoriasis. The competitive endogenous RNA (ceRNA) hypothesis posits that lncRNAs can not only regulate the expression of target genes directly but also influence the total number of miRNA molecules by binding to the core sequences of miRNAs and increasing the levels of target genes further affecting mRNAs. ceRNA is extensively involved in important biological activities [12]. The lncRNA-miRNA-mRNA network is constructed based on ceRNA theory [13]. Circular RNAs are evolutionarily conserved circular noncoding RNAs that play a role in numerous diseases by sequestering microRNAs (miRNAs). However, their role in psoriasis remains unclear. Since the discovery of non-coding RNAs, studies have revealed that many ncRNAs, such as microRNAs (miRNAs) and lncRNAs, are involved in the pathogenesis and evolution of the molecular pathology of psoriasis [14,15].

2- Problem Definition

Since the discovery of non-coding RNAs (ncRNAs), studies have revealed that many ncRNAs such as microRNAs (miRNAs) and long non-coding RNAs (lncRNAs) have long been involved in the pathogenesis and development of the molecular pathology of psoriasis [84,85]. However, the expression profile of another class of ncRNAs, circular RNAs (circRNAs), and their potential biological functions in psoriasis remain unclear. CircRNAs generated by alternative splicing represents a particular class of ncRNA molecules [86,87]. In contrast to linear RNAs, circRNAs are characterized by a covalently closed loop structure with neither 5- to 3- polarity nor a polyadenylated tail. CircRNAs are resistant to RNA exonuclease, which improves their stability [86].

lncRNAs are important regulators of the differentiation and activation of immune cells. They are involved in the development of autoimmunity and autoimmune diseases. Targeting lncRNAs may be beneficial for treating autoimmune diseases [88].

The challenge of my study is to perform a detailed exploratory analysis of lncRNA and circRNA

mediated ceRNA networks in psoriasis. The ceRNA network is used to identify lncRNAs and circRNAs that can function as competitive endogenous RNA networks to study the mechanistic pathways involved in the molecular pathogenesis of psoriasis. The terminology referred to as the ceRNAs (competing endogenous RNAs) network is used to describe the role of non-coding RNAs.

3- Problem motivation

The inspiration for my research problem is the ambiguous etiology and intricate pathogenesis of psoriasis, it remains incurable. Hence, there is a great requirement in the comprehension of molecular networks involved in psoriasis to boost the quality of psoriatic care. The research problem is interesting because a theory is, how ncRNAs work has been proposed and progressively confirmed as it is based on a novel ceRNA network methodology, which could be tested and applied in the application area, which is psoriasis.

4-Overview of previous research

RNAs have important roles in immune and inflammatory pathways by regulating gene expression through multiple mechanisms [35]. Among these mechanisms, the ceRNA theory is commonly studied. According to this theory, lncRNAs act as sponges for miRNAs and decrease the effects of miRNAs on their target genes, thus promoting the expression of target genes [36]. It is currently recognized that the pathogenesis of psoriasis is associated with the interleukin (IL)-23/type 17 T-helper cell immune axis, leading to abnormalities in immune cells and associated cytokines, which further induces excessive proliferation of keratinocytes[77].Cytokine-targeting drugs have been developed, including adalimumab (which targets tumor necrosis factor-α), secukinumab (which targets IL-17A), and ustekinumab (which targets IL-12 and IL-23 p40) [78].

lncRNA-MSX2P1 activates S100A71 and facilitates the growth of IL-22-stimulated keratinocytes by inhibiting miR-6731-5p [79]. A research study demonstrated that lncRNA H19 regulates the differentiation of keratinocytes by increasing desmoglein 1 expression through sponging miR-130b-3p [80].

CircRNAs have potential applications as diagnostic biomarkers and novel therapeutic targets for skin diseases due to their structural and functional properties. It is a special biomarker of skin disease [81]. Hsa_circ_0004287 inhibits M1 macrophage activation in an m^6A-dependent manner in psoriasis and may serve as a general therapeutic candidate for psoriasis [82]. However, studies on the role and mechanism of ceRNA in psoriasis are still lacking. The purpose of this study is to

use bioinformatic methods to construct and analyze the lncRNA-miRNA-mRNA-ceRNA network and the circRNA-miRNA-mRNA-ceRNA network in psoriasis. These results can provide useful information regarding the diagnosis and potential therapeutic targets for psoriasis. In relation to this, the ceRNA is a novel technique, however, it is not only being studied in cancers, but has also been studied in non-cancers such as lumbar disc degeneration and aortic dissection [37] [38].

5-Research aim and objectives

The current aim of this study is to identify lncRNAs and circRNAs that can act as competitive endogenous RNA and thus could play a crucial role in the pathogenesis of psoriatic disease by exploring the ceRNA networking using different bioinformatic tools. Significant ceRNA biomarkers for this disease are diagnosed using microarray data analysis.

The purpose of this study was to investigate at how the expression profiles of circular RNA (circRNA), long non-coding RNA (lncRNA), microRNA (miRNA/miR), and mRNA were altered in psoriasis by implementing differential gene expression and network analysis.

The objectives of this investigation are the following:

- Selecting and adopting a bioinformatics pipeline for differential expression evaluation of microarray data.
- Using web tools and databases for target prediction analysis.
- Creating a biological network for the ceRNA mechanism in cytoscape visualization software.
- Using cytoscape plug-ins to identify hub nodes for lncRNA and circRNA.
- Protein-protein interaction analysis and hub gene identification.

7-Potential scientific contribution

This research project aims at the computational identification of various lncRNAs and circRNAs that could be a therapeutic target for psoriasis. Most methodology steps have more than one tool or package. So, through this work, a comparative effectiveness of these tools could be seen, which will benefit scientists to determine which tools or packages to use for this type of research work, mRNA-miRNA target prediction was performed using the multiMiR and miRNATap R packages. These interactions represented regulatory relationships in the coding-non-coding RNA network, and multiple interactions for specific genes suggested a complex regulatory network. Next-generation methods were applied in this study, which yield promising results using various bioinformatics tools such as read oligo package and normalization with RMA package, differentially expressed genes with limma packages. Target prediction by used miRDip v4.1 followed by cytoscape and miRNA path tools. This study is augmented with several methods, experimental tools and biological data that could provide a good framework for treating the disease pathogenesis. constructed ceRNA network. All involved RNAs are associated with prognosis. This study will help scientists solve the biological challenge of psoriasis, but also help researchers understand the ceRNA network and possible biomarker of psoriasis.

8 - METHOD DESCRIPTION

8.1- Gene expression omnibus (GEO) dataset collection

A microarray dataset was retrieved from the GEO public database (www.ncbi.nlm.nih.gov/geo); The accession number of the data set used is GSE145305, which contains two sub-data sets: GPL17586, [HTA-2_0] Affymetrix Human Transcriptome Array 2.0 [transcript (gene) version] and GPL19117[miRNA-4] Affymetrix Multispecies miRNA-4 Array. The microarray dataset GSE145305 is downloaded and extracted in ZIP format. Some important descriptions are observed for additional processes such as platform ID and annotation details.

The GSE145305 dataset is used here for combined miRNA-mRNA transcription profiling in human skin. So, this has the data for both miRNAs and mRNAs which is the best thing to do this research. The platform organism is Homo sapiens and has two subgroups with 14 samples, the 8 samples are miRNA, of which 4 are psoriasis skin samples and 4 are healthy skin samples, and 6 samples of mRNA, of which 3 are psoriatic skin samples and 3 are healthy skin samples.

The research used the NCBI databases. The NCBI abbreviation is: National Center for Biotechnology Information provides a wide variety of online biological database resources, including the GenBank Nucleic Acid Sequence Database and the PubMed Database. PubMed database is famous for citations and abstracts published in life science journals [75].

8.2-Microarray data analysis

Microarrays provide data advantageous to study the activity of all genes present in an organism through log2 transformation. Microarray data analysis has been one of the most important achievements in the interaction between statistics and bioinformatics [83].

This study has used NCBI GEO because this archive handles both treated and untreated data and is the largest fully public resource on gene expression. The raw data were obtained from microarray datasets. The raw data for both mRNA and miRNA are in the format of CEL files. Then oligo package was used to read it in an R environment. A package for analyzing oligonucleotide arrays at the probe level is used as a tool analysis of Affymetrix gene expression arrays [19][20].

Data normalization is performed to remove the variations and get high quality data, which is performed with the Robust Multiarray Average (RMA) method using bioconductor. RMA is used for

normalization because it performs probe-level normalization, potentially reducing the breakdown of information that corrects for background noise in the data. It effectively protects the outliers as it corrects the data biases using quantile normalization [75].

Then, normalized with the RMA package. RMA has become a widely used method for preprocessing Affymetrix gene expression microarrays. The software is implemented in R packages and consists of three steps: background correction, quantile normalization, and expression calculations. A limitation of RMA is that the calculated probe set intensities change when a set of microarrays is re-preprocessed after adding additional microarrays to the analysis set [21].

The limma package is used for differential expression analysis. It stands for linear model of microarray analysis. Limma is the R/Bioconductor software package that provides the solution for analyzing data from gene expression experiments. The limma package is a core part of the bioconductor used in statistical analysis of the genomic data. Package design consists of three steps: first, initial preprocessing, filtering, and then normalization.

In limma, the number of statistical principles is performed and is also suitable for a linear model. The advanced statistical algorithm in limma enables information borrowing, quantitative weighting, variance modeling and data pre-processing [22][23]. The limma package contains statistical methods that facilitate information borrowing using empirical Bayesian methods to obtain later estimates of variance, incorporate observation weights to account for variations in data quality, allow variance modeling to account for technical or biological heterogeneity and pre-processing methods such as variance stabilization to reduce noise. These methods all contribute to improving inference at both the gene and gene set level in small experiments.

Limma integrates several statistical principles in a way that is effective for large-scale expression studies. It works with a matrix of expression values, with each row representing a gene or other genomic trait relevant to the current study, and each column corresponding to an RNA sample. On the one hand, it fits a linear model to each row of data and uses the flexibility of such models in various ways, for example to handle complex experimental designs and to test very flexible hypothesis. On the other hand, it exploits the highly parallel nature of genomic data to borrow strength between the gene-by-gene models, allow for different degrees of variability between genes and between samples, and make statistical inferences more reliable when sample numbers are small. All functions of the statistical models can be accessed not only for genetic expression analysis but also for higher-level analysis of gene expression signatures [89].

8.3- mRNA-miRNA target prediction

It is very important to define target prediction of miRNAs for many reasons. Defining the target

set of a miRNA is key for many biologists in understanding its biological role. For bioinformaticians developing miRNA-validated targets, they offer the best biomarkers. The identification of miRNA targets has followed three general approaches: bioinformatic target prediction, biochemical isolation of miRNA/mRNA complexes, for transcriptome/proteome analysis. MiRNA are the short non-coding RNAs that serve as regulators of transcriptional and post-transcriptional gene expression. They regulate their target messenger RNAs (mRNA) through their binding and inactivation. Even in this modern era and high-throughput technologies, it is difficult and abundant to detect all possible microRNA targets, and even bigger problem is that there are many false positives so there are many tools available, one of which is the miRNAtap. After differential expression analysis, used the miRNAtap [24], miRNAtap is the R package for aggregated microRNA target prediction.

- It facilitates the implementation of workflows requiring miRNA predictions.
- It allows the integration of ranked miRNA target prediction.
- It improves the quality of predictions above any of the single sources.

Many computational microRNAs target prediction tools have been developed, each with its strengths and weaknesses. The integration of different tools was a successful approach to minimize the shortcomings of individual databases. Here used miRDip v4.1 which provides nearly 152 million human microRNA target predictions collected across 30 different resources [25][26].

8.4- miRNA-lncRNA target prediction

LncRNA plays a crucial role in autoimmune infections and immune-related diseases. lncRNAs are >200 nucleotides, they cannot encode a protein, but they can be expressed as alternatively spliced variants [27]. In addition, lncRNAs fulfill important functional roles in epigenetic, transcriptional, or post-transcriptional regulation [28]. LncRNAs play a crucial role in the regulation of immune-mediated diseases and contribute to the pathogenesis of psoriasis by regulating the expression of protein-encoding genes [29, 30].

LncBase and starBase are used to find out the interactions between differentially expressed miRNAs and target lncRNA [31,32]. The purpose of using DIANA-lncBase is to assist researchers in experiments and to unravel putative functional interactions of microRNA (miRNA)-lncRNA [33]. LncBase aims to become a reference repository for lncRNA-miRNA interactions through novel computational approaches [34].

The DIANA-LncBase pipeline flow is the best available lncRNA resource collected for human and mouse species. The steps included in the pipeline flow are lnc transcript filtering, lnc categorization, experimental MRE identification, in silico target prediction, tissue expression, evolutionary preservation and finally database integration [33][35].
StarBase is a database for miRNA-mRNA map interactions. MicroRNAs (miRNAs), other non-coding RNAs (ncRNAs) (e.g., lncRNAs, pseudogenes, and circRNAs), and competing endogenous RNAs (ceRNAs) have and in various human diseases unpredictably little knowledge of the regulatory interaction networks among the multiple classes of RNAs. StarBase is the database that systematically identifies the RNA and protein interaction networks from 108 CLIP-Seq datasets. The Clip-Seq dataset is used to find out which RNA interacts with a particular RNA-binding protein. The starBase database shows:

- ~9000 miRNA-circRNA.
- 16 000 miRNA-pseudogene.
- 285 000 regulatory relationships between protein–RNA.

In addition, starBase has been updated to provide the most comprehensive CLIP-Seq experimentally supported miRNA-mRNA and miRNA-lncRNA interaction networks to date. It also identified 10,000 ceRNA pairs of CLIP-assisted miRNA target sites [32[36]. It is an open-source program for studying miRNA-ncRNA, miRNA-mRNA, ncRNA-RNA, RNA-RNA, RBP-ncRNA, and RBP-mRNA interactions from CLIP-seq, Degradome-seq, and RNA interactome data [33].

8.5- miRNA-circRNA target prediction

CircBase and starBase are used to find out the interaction between differentially expressed miRNAs and circRNAs. CircBase is the database for the interaction of miRNA-circRNA [29]. The main features of circBase are [37,38]: it provides users with a single repository that collects, unifies, and annotates circRNA data in standardized formats. It extends public datasets with additional fundamental analysis. It integrates this data with external resources, such as UCSC Genome Browser and NCBI databases. It allows users to interact with the database contents through a simple interface.

8.6- Construction of a combined ceRNA network using mRNA-miRNA-lncRNA and mRNA-miRNA-circRNA

Cytoscape is a general-purpose open-source software environment for extensive integration and easy installation. It is developed in the Java programming language. It handles basic functions like network layout and the mapping of data attributes to visual display properties. The interactions were combined in the cytoscape software to build a network in which the following interactions coexist; mRNA-miRNA-lncRNA and mRNA-miRNA-circRNA. This network will be our ceRNA network. The ceRNA hypothesis suggests that RNAs can regulate each other by competing for a limited pool of miRNAs [39,40]. The tool used for the ceRNA network in cytoscape [40]. It offers large-scale integration of data from molecular interaction networks. The general feature of cytoscape is:

- Data visualization (to visualize and integrate the network).
- Network analysis (basic functionality to layout and query the network).

8.7- Hub lncRNA and Hub circRNA identification

Cytoscape is an open platform with many plugins to extend either way one of the visualization options and the power of network analysis. CytoHubba is one of cystoscope's advanced plugins. The cytoscape plugin cytoHubba offers 11 topological analysis methods. The dense ceRNA network is analyzed with the cytoHubba plugin [40,43]. CytoHubba offers an easy-to-use interface for exploring important nodes. It helps in identifying hub objects and sub-networks from the complex interactome.

There is the shortest way cytoHubba has 11 topological methods, and one method is MCC [41]. MCC stands for Maximum Clique Centrality. It's a more effective way to find hub nodes. It has different properties because it gives more accurate results, is more reliable in unbalanced data and gives high results [44]. The connectivity component is used to sort strongly connected nodes. The highly crosslinked lncRNAs and circRNAs are considered hub lncRNAs and hub circRNAs.

8.8- Protein-Protein Interaction (PPI) analysis of genes presents in the ceRNA network.

For the PPI network used the STRING database. It is used for protein-protein association data for many organisms. The aim is to collect and integrate this information. The STRING database includes both direct and indirect association types, i. e. physical and functional association [45].

The mRNA nodes present in the ceRNA network are extracted and subjected to protein-protein interaction analysis using the STRING database. STRING is the database of functional associations [46]. The network was visualized in cytoscape software.

8.9- Hub gene module identification

MCODE is a cytoscape plugin that finds clusters (strongly interconnected regions) on a network. Clusters mean different things in different types of networks. The PPI network analyzed an MCODE plug-in, MCODE is used to identify highly connected modules. MCODE is a relatively fast clustering method in relation to computing resources. The main features of the MCODE are fast network clustering, fine tuning of results and cluster finding parameters. It also supports interactive cluster boundary and content exploration with multiple techniques for managing result sets. The main disadvantage is that it does not support statistical result evaluation. Creation of cluster subnetworks and export in plain text; these modules are called hub gene modules. The genes present in this module

could be of great importance as biomarkers for the disease.

The hub genes will be subjected to a pathway analysis using R packages clueGO to find out different biological processes, cellular components, and molecular functions of those genes.

9- Implementation and Results

9.1 -Microarray data pre- processing and differential expression analysis

Raw microarray data was downloaded from NCBI GEO. GSE145305 was downloaded for mRNA and miRNA analysis. The sample detail is shown in Table 1. The raw data was in the format of a .CEL file. Microarray data were analyzed in R. The raw data was read with oligo R-Package. The data were normalized using the RMA function of the oligo package. RMA has become a widely used method for preprocessing Affymetrix gene expression microarrays. RMA packages consist of three steps: background correction, quantile normalization, and expression calculations. Differential expression analysis was performed using the limma R package.

The p-value threshold was less than 0.05 to be significant for both mRNA and miRNA. For mRNA, the LogFC threshold is greater than 2 for upregulated genes and less than -2 for downregulated genes. For miRNA, the LogFC threshold is greater than 1.5 for upregulated genes and less than -1.5 for downregulated genes. In the case of mRNA, 10 genes were significantly down-regulated and 86 significantly up-regulated. And in the case of miRNA, 48 were significantly down-regulated and 75 were significantly up-regulated.

Table 1: - Microarray Platform and Sample details

Platform ID	Platform Name	Samples (PS)	Samples (HS)
GPL17586	[HTA-2_0] Affymetrix Human Transcriptome Array 2.0 [transcript (gene) version]	GSM4314529 GSM4314530 GSM4314531	GSM4314532 GSM4314533 GSM4314532
GPL19117	miRNA-4] Affymetrix Multispecies miRNA-4Array	GSM4305469 GSM4305470 GSM4305471 GSM4305472	GSM4305473 GSM4305474 GSM4305475 GSM4305476

PS (Psoriasis Skin), HS (Healthy Skin)

One of the most important characteristics of data analysis is exploratory data analysis. It involves using statistical techniques to identify patterns that may be hidden within a group of numbers. One of the techniques is the box plot, which is used to visually summarize and compare groups of data. In my research study I used the two datasets, one of the mRNAs and one of the miRNAs. The samples of each study are 8 and 6, respectively, and the boxplot uses the median, approximate quartiles, and lowest and highest data points to convey the level, spread, and symmetry of a distribution of data values. It can also be easily refined to identify outlier data values and easily constructed by hand. The boxplot, like other visual methods, is a tool that can improve our reasoning about quantitative information. It also provides information about the outliers [51].

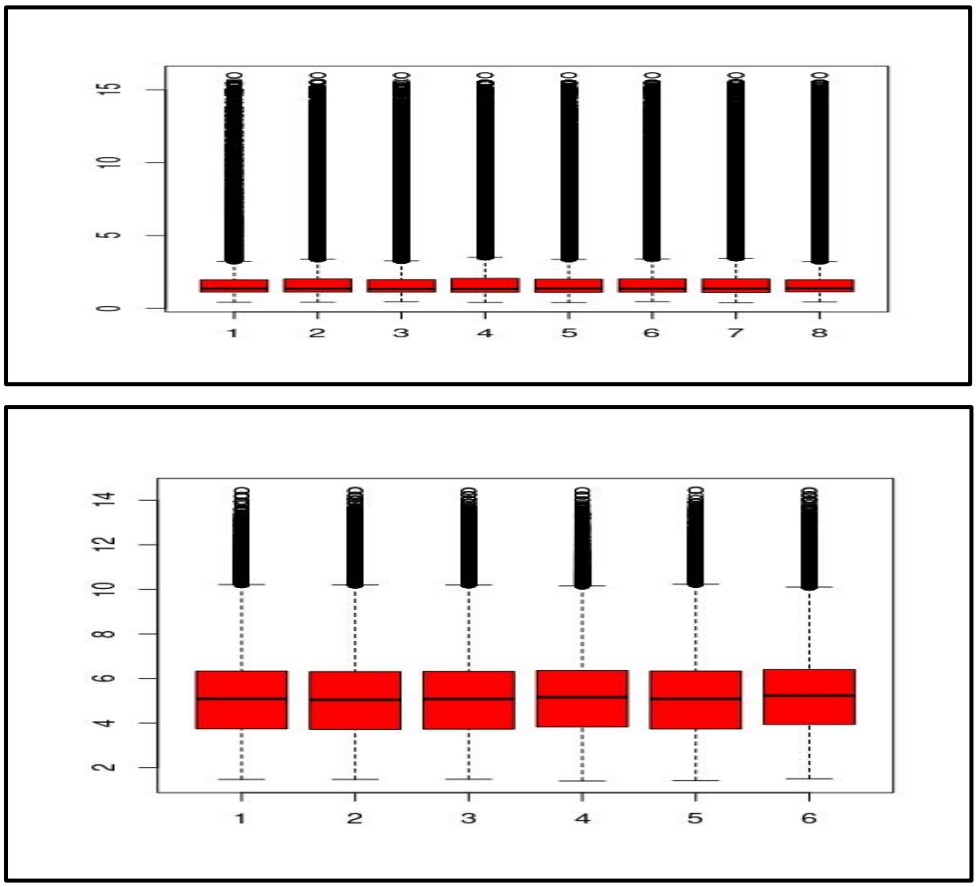

Figure 1: The expression box plot after normalization of GSE145305_miRNA and GSE145305_mRNA. The black line inside the box of the median of the expression is almost a straight line in the boxplot after normalization, indicating a high level of normalization.

9.2-Differential expression and volcano plots

One of the most common tasks to perform by using microarrays is the analysis of differential expression, such as the identification of mRNAs/genes, miRNA, whose expression level is different under two circumstances, i.e. in the diseased/healthy, before/after condition. Differentially expressed genes provide insight into the biological processes involved in disease. They can be used as biomarkers to analyze and predict diseases. The term differential expression means that the average expression level of an mRNA and miRNA in each study sample is much greater or less than that in another study sample group [52]. The volcano plot most referred to as the scatterplot with log10p-value, the t-test as the y-axis and (log10) FC as the x-axis. It allows rapid visual identification of genes with large fold changes that are also statistically significant. These may be the most biologically significant genes. In a volcano diagram, the most upregulated genes are to the right, the most downregulated genes to the left, and the most statistically significant genes to the top [53].

After using limma, the p-value threshold is assumed to be less than 0.05 for both mRNA and miRNA as this is considered significant. For mRNA, the logFC threshold is greater than 2 for upregulated genes and less than -2 for downregulated genes. For miRNA, the logFC threshold is greater than 1.5 for upregulated genes and less than -1.5 for downregulated genes.

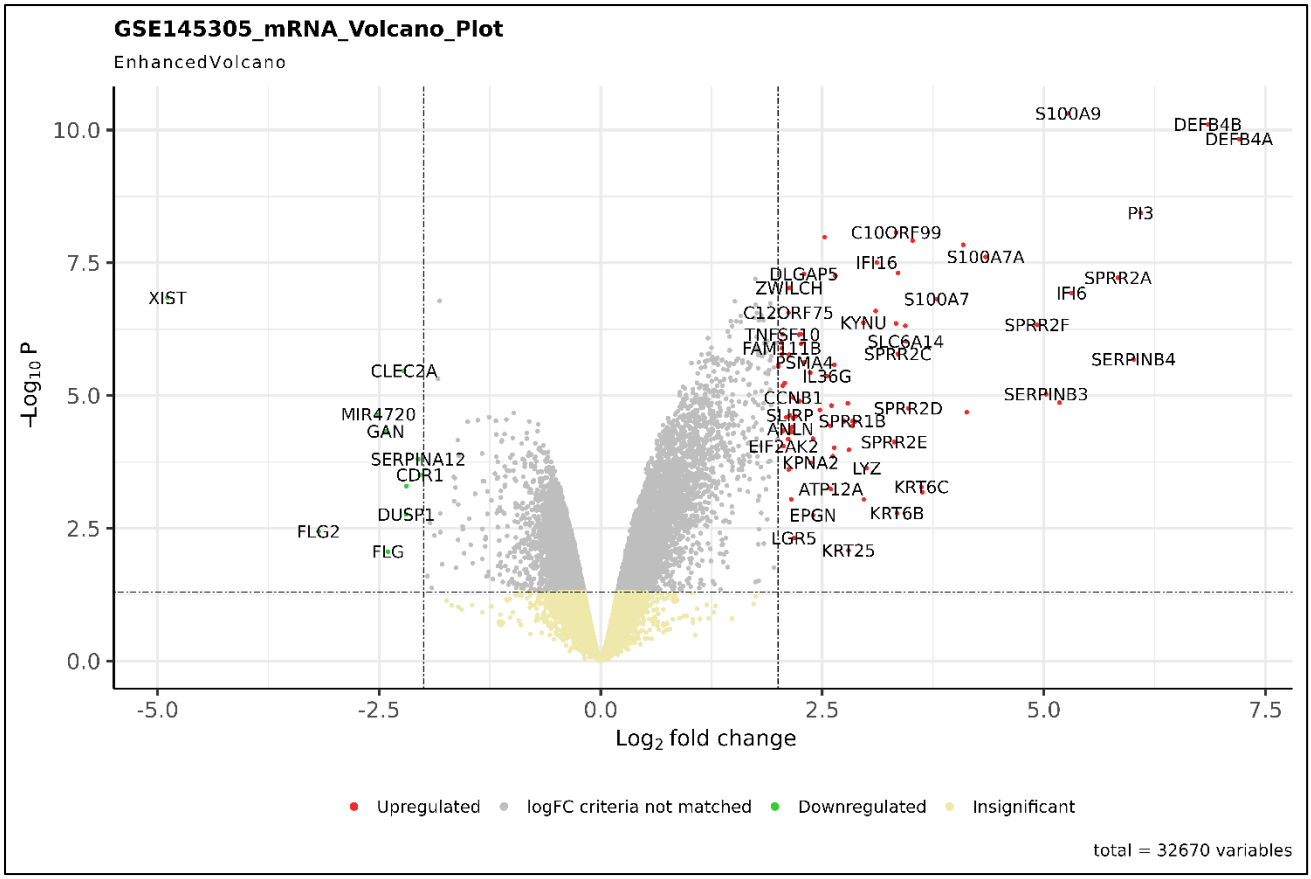

Figure 2: Volcano plot of dataset GSE145305_mRNA. It shows the differential expression of genes and major genes. For mRNA, the logFC threshold is greater than 2 for upregulated genes and less than -2 for downregulated genes. The green dot shows the down-regulated genes and the red dot the up-regulated genes and at the top the most important genes.

Table 2: Top Up-regulated and down-regulated genes of GSE145305_mRNA

Regulation	Gene	P-Value	LogFC
Down	FLG	0.0086662	-2.40099898
Down	FLG2	0.0036218	-3.18388711
Down	DUSP1	0.0017313	-2.19604427
Down	XIST	0.0000001	-4.89008573
Down	CDR1	0.0003109	-2.04010919
Up	IFITM4P	0.0000460	2.05767
Up	CNN	0.0008919	2.14964
Up	DSC2	0.0000954	2.63495
Up	SERPINB3	0.0000094	5.02619
Up	SERPINB4	0.0000020	6.01307

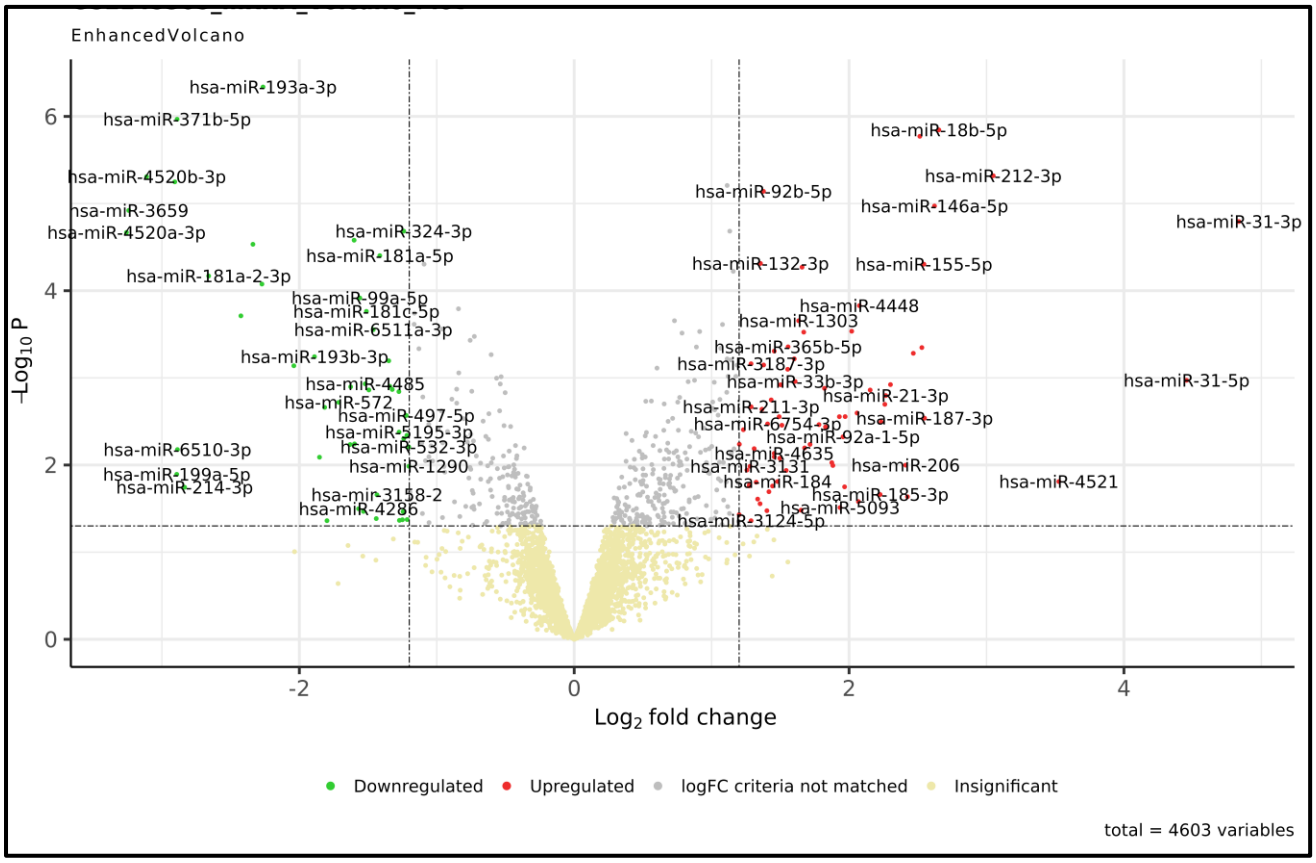

Figure 3: Volcano plot of data set GSE145305_miRNA. It shows the differential expression of genes and major genes. For miRNA, the logFC threshold for upregulated genes is greater than 1.5 and less than -1.5.

Table 3: Top Up-regulated and down-regulated genes of GSE145305_miRNA

Regulation	Gene	P-Value	LogFC
Down	hsa-miR-196b-3p	0.00063	-1.3495
Down	hsa-miR-181a-5p	0.02196	-1.4352
Down	hsa-miR-4708-3p	0.0424	-1.7989
Down	hsa-miR-572	0.0435	-1.56019
Down	hsa-miR-4667-5p	0.00172	-2.89216
Up	hsa-miR-31-3p	0.000016	4.836
Up	hsa-miR-31-5p	0.001067	3.6345
Up	hsa-miR-4521	0.015485	3.5264
Up	hsa-miR-212-3p	0.000004	3.0491
Up	hsa-miR-18b-5p	0.000001	2.6532

9.3- mRNA-miRNA target prediction

Since mRNA and miRNA are oppositely expressed, this means that upregulated miRNA can downregulate genes and downregulated miRNA can lead to upregulation of genes. Thus, only oppositely expressed mRNA-miRNA pairs were taken for mRNA-miRNA target prediction. Therefore, two lists were created using up-regulated mRNA-down-regulated miRNA and down-regulated mRNA-up-regulated miRNA. An R-package miRNATap'' and an online tool miRDip were used to perform target prediction of mRNA-miRNA pairs. 14 pairs were identified for target prediction analysis of downregulated mRNA and upregulated miRNA, and 65 pairs were identified for target prediction analysis of upregulated mRNA and downregulated miRNA. The interaction was visualized in cytoscape.

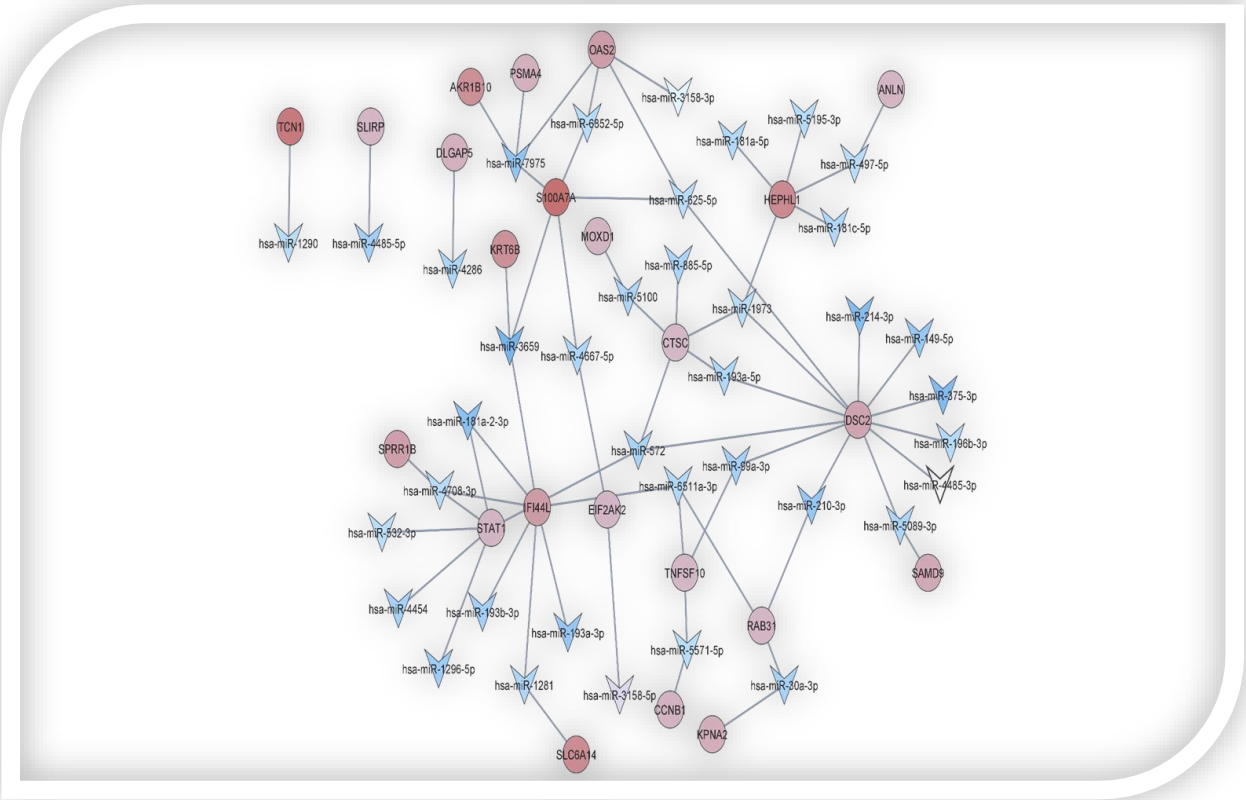

Figure 4: Circular nodes are mRNAs and arrow-shaped nodes are miRNAs. The color intensity represents the degree of up- or down-regulation. Red color shows up regulation and blue color down regulation.

Table 4: mRNA-miRNA target prediction.

	Gene Symbol	MicroRNA		Gene Symbol	MicroRNA
1)	AKR1B10	hsa-miR-7975	**8)**	SLC6A14	hsa-miR-1281
2)	CTSC	hsa-miR-193a-5p		S100A7A	hsa-miR-3659 hsa-
		hsa-miR-1973			miR-4667-5phsa-
		hsa-miR-1973			miR-625-5p hsa-miR-
		hsa-miR-572			6852-5p
		hsa-miR-885-5p			hsa-miR-7975
3)	ANLN	hsa-miR-497-5p	**9)**	SLIRP	hsa-miR-4485-5p
4)	CCNB1	hsa-miR-5571-5p	**10)**	SPRR1B	hsa-miR-4708-3p
	DSC2	has-miR-149-5p		STAT1	hsa-miR-1296-5p
		hsa-miR-193a-5p			hsa-miR-4454 hsa-
		hsa-miR-196b-3p			miR-4708-3phsa-
		hsa-miR-1973 hsa-			miR-532-3p hsa-
		miR-210-3p hsa-miR-			miR-572
		214-3p hsa-miR-375-			
		3p hsa-miR-4485-3p			
		hsa-miR-5089-3p			
		hsa-miR-572			
		hsa-miR-625-5p			
		hsa-miR-99a-3p			
6)	EIF2AK2	hsa-miR-3158-5p	**11)**	TNFSF10	hsa-miR-5571-5p hsa-
		hsa-miR-4667-5p			miR-6511a-3p

7)	HEPHL1	hsa-miR-181a-5phsa-miR-181c-5phsa-miR-1973 hsa-miR-497-5p hsa-miR-5195-3p	**12)**	IFI44L	hsa-miR-1281 hsa-miR-181a-2-3p hsa-miR-193a-3p hsa-miR-193b-3p hsa-miR-3659
8)	DLGAP5	hsa-miR-4286			

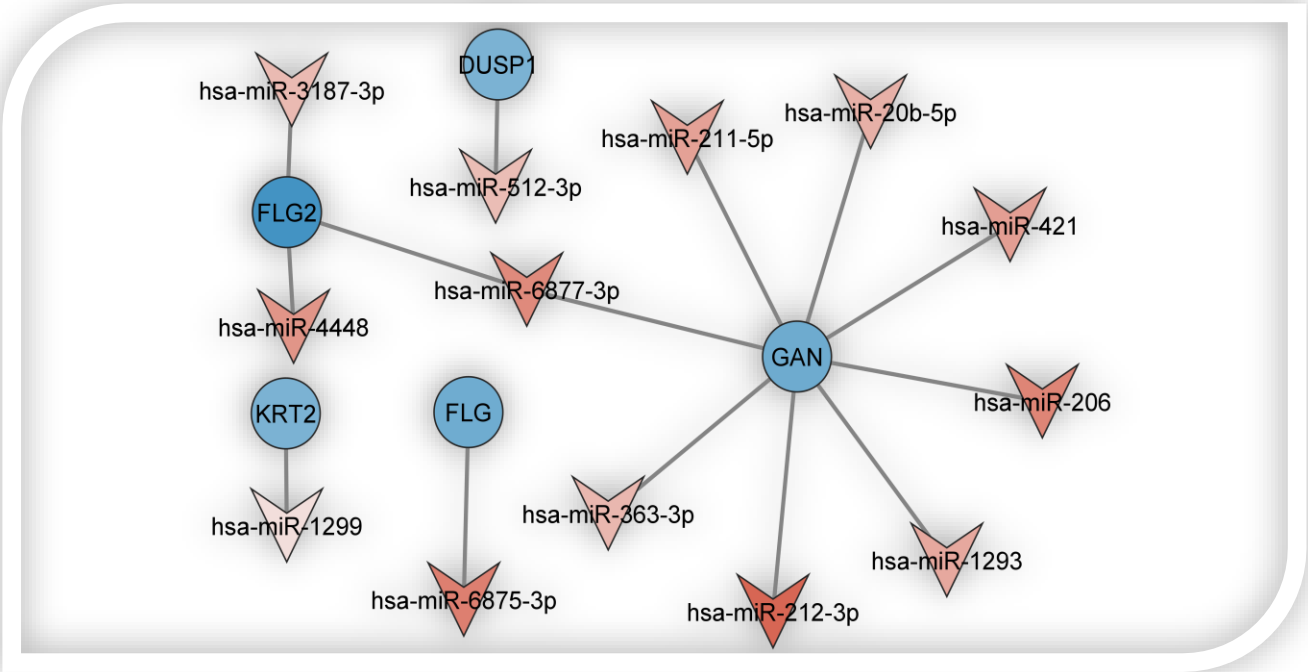

Figure 5: Circular nodes are mRNAs and arrow-shaped nodes are miRNAs.
The color intensity represents the degree of up- or down-regulation. Red color shows up regulation and blue color down regulation.

Table 5: downregulated -mRNA-upregulated _miRNA

	Gene Symbol	Micro RNA
1	DUSP1	hsa-miR-512-3p
2	FLG	hsa-miR-6875-3p
3	FLG2	hsa-miR-3187-3p
4	FLG2	hsa-miR-4448
5	FLG2	hsa-miR-6877-3p
6	GAN	hsa-miR-1293
7	GAN	hsa-miR-206
8	GAN	hsa-miR-20b-5p
9	GAN	hsa-miR-211-5p
10	GAN	hsa-miR-212-3p
11	GAN	hsa-miR-363-3p
12	GAN	hsa-miR-421
13	GAN	hsa-miR-6877-3p
14	KRT2	hsa-miR-129

9.4- miRNA-lncRNA target prediction and miRNA-circRNA target prediction.

StarBase (https://starbase.sysu.edu.cn/) was used. The input was the miRNA list from the mRNA-miRNA target prediction (both up and down). In the selection process on the website, the CLIP data was determined to be high stringency and the degradome data to be low stringency. Stringency is the parameter setting. It described the combination of conditions under which the probe's target is exposed. The number of supporting CLIP-seq experiments and degradome-seq data can be adjusted to control the stringency of the predictions. CLIP supported miRNA target sites.

9.5- Construction of combined ceRNA network using mRNA-miRNA-lncRNA and mRNA-miRNA-circRNA.

The ceRNA hypothesis has been proposed, postulating that RNAs can interact with each other and manipulate biological functions independently of protein translation [47]. There are two ceRNA networks in this research. A network can contain different types of ceRNAs, such as lncRNAs, circRNAs, pseudogenes and mRNAs-lncRNA-mediated ceRNA network consists of mRNA, miRNA and lncRNA. circRNA-mediated ceRNA network consists of mRNA, miRNA and circRNA. A regression analysis was performed between lncRNAs, mRNAs, circRNA and miRNA, the gene with the highest level of connectivity in the ceRNA network. Cytoscape software was used to construct the networks.

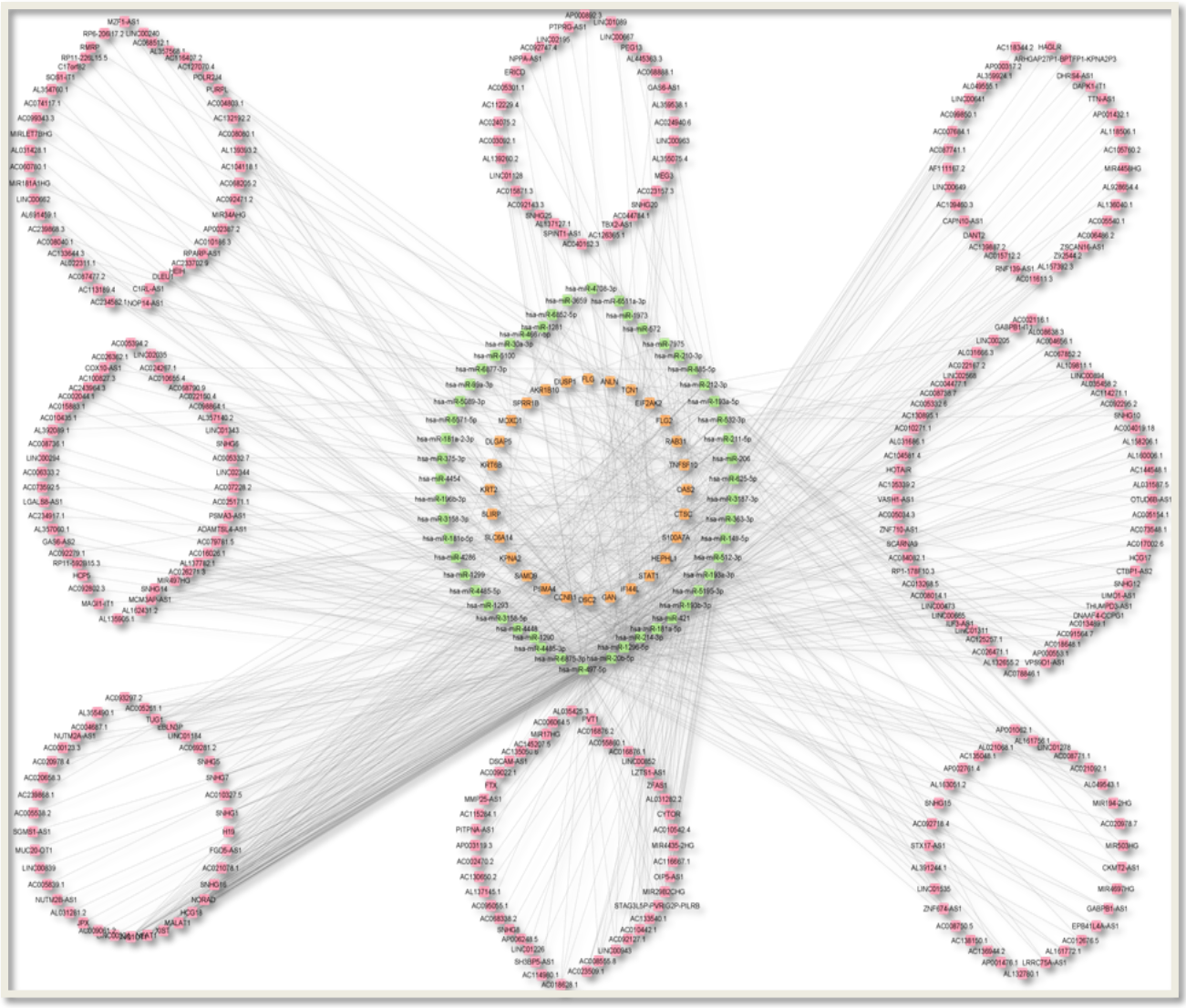

Figure 6: The ncRNA-mediated ceRNA network consists of mRNA, miRNA, and lncRNA.

For lncRNA-miRNA-mRNA, pink nodes are lncRNA, green nodes are miRNA, and orange nodes are mRNA. ceRNA network shows detailed description and crosstalk of all RNAs.

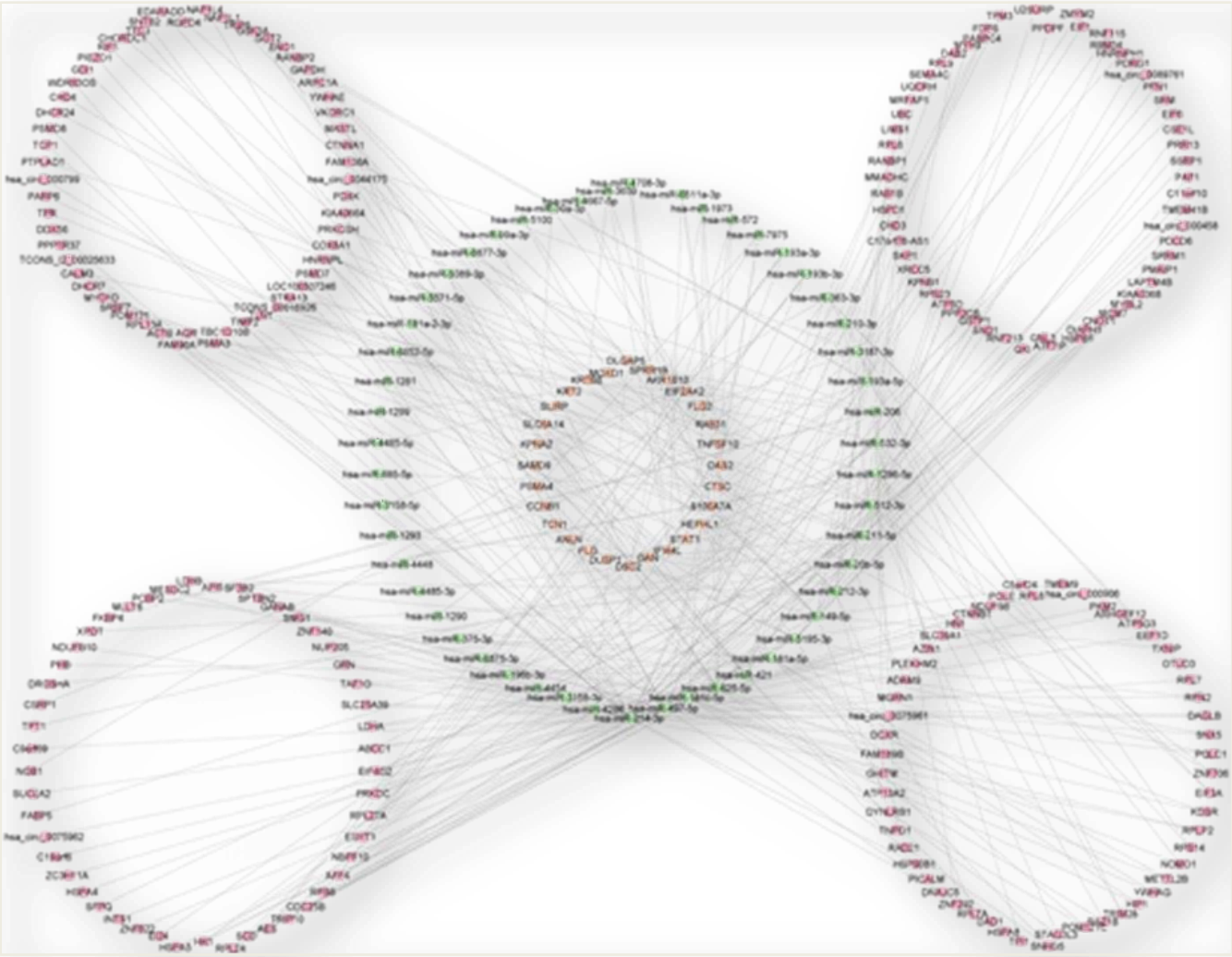

Figure 7: The circRNA-mediated ceRNA network consists of mRNA, miRNA, and circRNA.
For circRNA-miRNA-mRNA, outer nodes are circRNA, middle nodes are miRNA, and inner nodes are mRNA. The ceRNA network shows the detailed description and crosstalk of all RNAs.

9.6- Hub lncRNA and hub circRNA identification

Hub-circRNA and Hub-lncRNA are entries that have many interactions with different miRNAs because the ceRNA is too confusing. It is better to find out the hub circRNA and the hub lncRNA. CytoHubba plugin was used to find the degree of nodes. A degree cut-off of greater than or equal to 7 was set for lncRNA and a degree cut-off of greater than or equal to 4 was set for circRNA. This is the parameter setting in the cytoHubba. After thresholding, Hub circRNA was found to be DSC2, GAN, HEPHL1, IFI44L, HK1, STAT1 and Hub lncRNA to be XIST, NEAT1, KCNQ1OT1, MALAT1, HCG18, NORAD.

Table:6 Hub lncRNA and Hub circRNA identification with node degree.

CircRNA	Degree	lncRNA	Degree
DSC2	7	XIST	7
GAN	6	NEAT1	6
HEPHL1	5	KCNQ1OT1	5
IFI44L	5	MALAT1	5
HK1	4	HCG18	4
STAT1	4	NORAD	4

9.7- Analysis of the protein-protein interaction (PPI) of genes present in the ceRNA network

Protein-protein interactions (PPIs) play important roles in biological processes, cell-to-cell interactions, and in metabolic and developmental control [48]. All mRNAs present in the ceRNA network were taken for a protein-protein interaction. Cytoscape was used to perform protein-protein interaction. Recent developments have also led to the construction of networks that all exhibit protein-protein interactions, using computational methods for signaling pathways and identification of protein complexes in specific diseases.

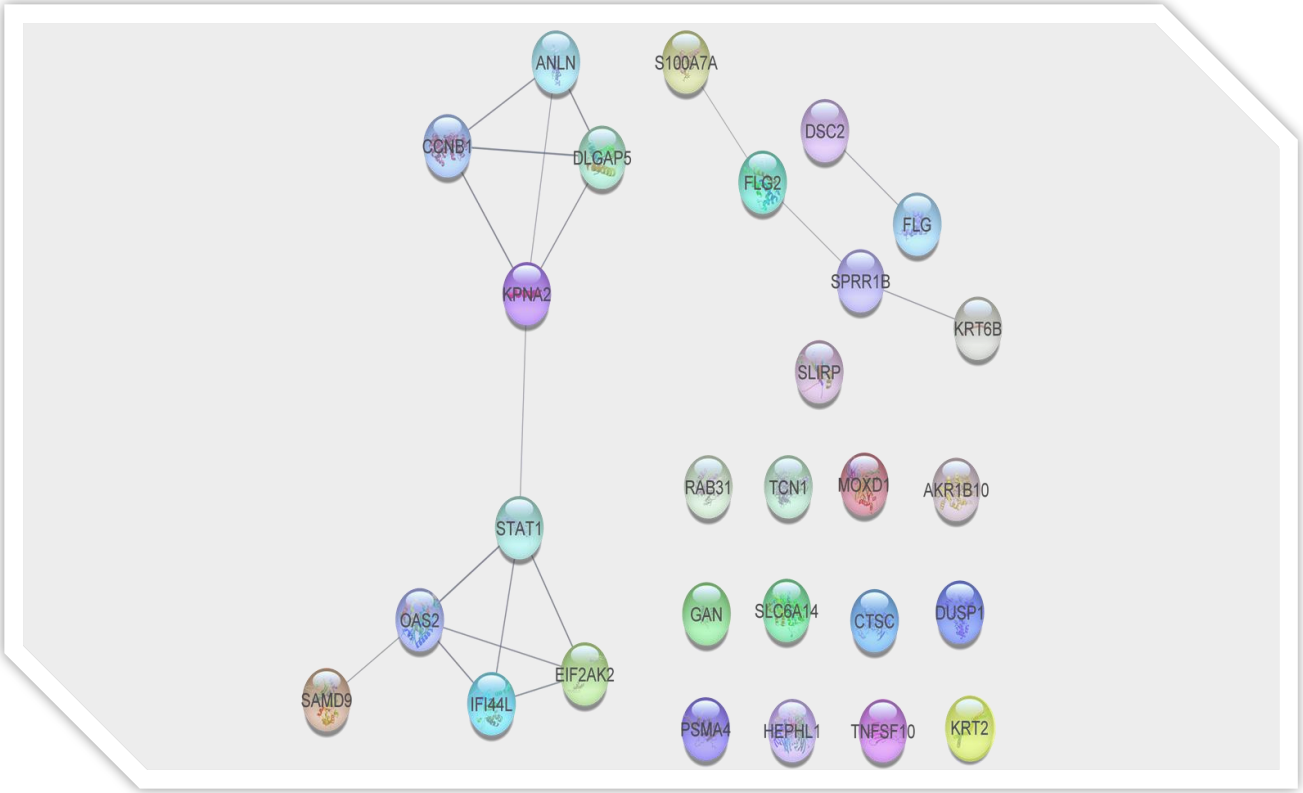

Figure 8: PPI protein-protein interaction present in ceRNA network. The most closely linked genes displayed by Cytoscape via the String plugin.

9.8- Hub gene module identification from then PPI network:

Finding protein-protein interaction nodes/graphs helps identify drug targets [49]. Previous studies have shown that proteins with a larger number of interactions (hubs) can include many factors such as enzyme families, transcription factors, and intrinsically disordered proteins [50, 57]. In my research I used another Cytoscape plugin MCODE to find the hub module from the PPI STRING network.

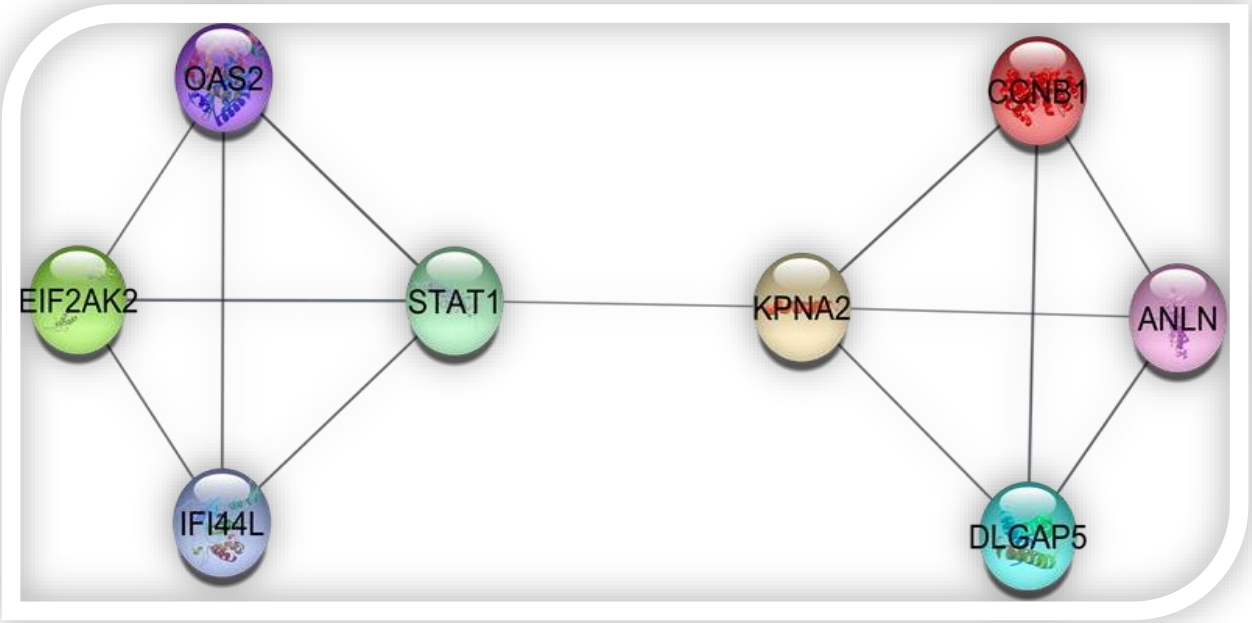

Figure 9- **Hub gene identification**: A total of 8 genes that are strongly linked.

9.9- Analysis of associated biological functions:

The mRNA nodes present in the ceRNA networks were used as input for this analysis. The cytoscape clueGO plugin was used. The biological process, the molecular function and the cellular component were selected. A P-value less than 0.05 was considered significant. An enrichment network was created. Regulation of sister chromatid segregation and regulation of cell cycle metaphase/anaphase transition were some of deregulation is a common factor in psoriasis. Sister chromatid segregation is a key the significantly improved signaling pathways for the selected genes. Cell cycle feature of mitosis. In the anaphase of mitosis, the sister chromatids are said to separate at opposite poles of the cell. If deregulation occurs at this stage, it could be very

damaging. Another key symptom of psoriasis is the keratinization of the skin cells. Cornification means the hardening of skin cells. The genes found from the ceRNA are associated with these psoriasis-related signaling pathways, suggesting that these genes can be used for further analysis. These deregulated genes could be a good target for psoriasis therapeutics.

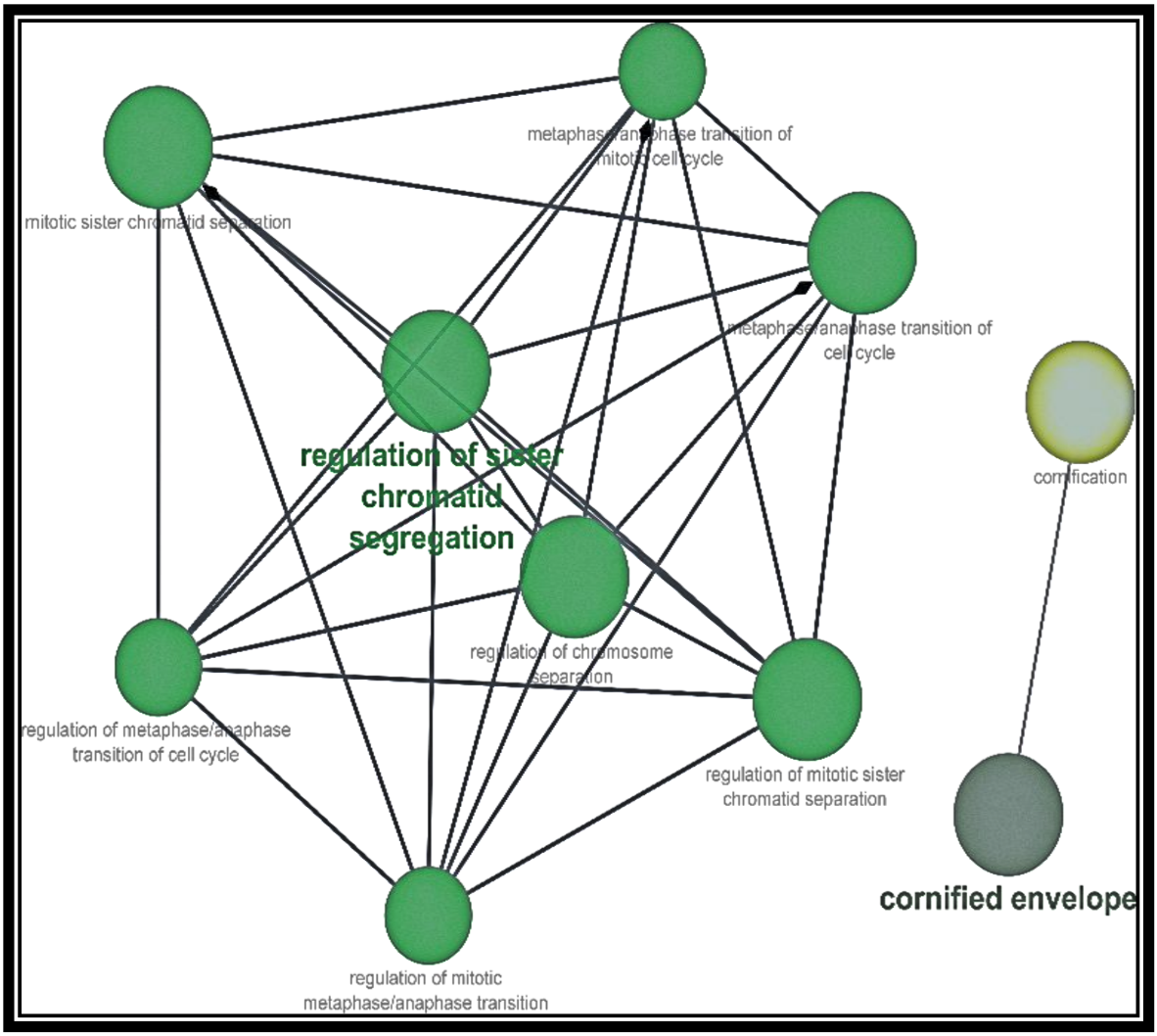

Figure 10: It shows associated biological functions. Cell cycle was one of the significantly improved signaling pathways for the selected genes. Cell cycle deregulation is a common factor in psoriasis

10- Discussion

Psoriasis is a skin disorder driven on by an immune reaction. The prime reason of psoriatic pestilence development is the interrelationships between the innate and adaptive immune responses [54]. Psoriasis pathophysiology research has substantially increased our knowledge of epidermal physiology. Psoriasis etiology advances have laid the groundwork for efficaciously tailored and personalized medications, offering significant insight into the pathophysiology of this chronic inflammatory skin illness. Though recent research has made significant progress in the treatment, prognosis, and diagnosis of psoriasis, the presence of psoriasis cannot be ruled out, possibly due to a lack of effective biomarkers and unexplained pathways that dictate psoriasis pathogenesis [55,56].
Recent research has shown that lncRNAs play an essential role in the development of several illnesses, and some of these lncRNAs could be used as indicators to enhance prognosis and diagnosis efficacy. lncRNAs serve a variety of purposes. circRNAs, in general, have a vital role in the prediction of certain malignancies [57]. The ceRNA hypothesis suggests that lncRNAs and circRNAs can participate with miRNAs for miRNA binding sites, affecting expression levels through MREs. This concept unifies the relationships between lncRNAs, circRNAs, mRNAs, and miRNAs, and adequately addresses the genetic association between various RNA forms [58]. Non-coding RNA has been recognized to exhibit a crucial function in a variety of disorders in various investigations. The competitive endogenous RNA (ceRNA)-mediated regulation pathways in psoriasis, on the other hand, are still unknown [57,58]. The purpose of this study was to investigate at how the expression profiles of circular RNA (circRNA), long non-coding RNA (lncRNA), microRNA (miRNA/miR), and mRNA were altered in psoriasis by implementing differential gene expression and network analysis.

10.1- Microarray data pre-processing

Microarray allows researchers to examine the expression of different genes. Microarray-based methodologies can be implemented to investigate the amount of generated mRNA with respect to the gene expression profile [59]. These approaches enable large-scale gene expression analysis at the same time. Microarray allows clinicians to analyze the expression and modulation of multiple genes at the same time, allowing them to detect modulated disease genes [60]. Pre-processing is perhaps the most crucial step in delivering correct data because it eliminates mistakes between arrays and identifies DEG's (Differentially Expressed Genes) between arrays and among an array. Pre-processing guarantees that comparable genes are manifested uniformly across arrays and among an array, even though the environmental circumstances induce them to be transcribed differently. The genes that

continue to express differently even after the pre-processing processes have been completed are considered as sick genes [61].

There are various strategies for pre-processing of the microarray data, such as MAS.04 and MAS.05, which are only suitable to single chips and rely on low-intensity MM bonding, which is ineffective in removing all disturbances. Although the UPC is a platform-independent tool, it can only assess the likelihood of gene expression. The RMA approach is preferable since it overcomes the drawbacks of the prior method. To correct background sounds and mistakes, RMA uses quantile normalization and PM intensity [62]. As a result, the RMA approach was used in the current investigation for the pre-processing of the microarray data.

10.2-Differential Gene Expression

The current investigation identified IFITM4P, CNN, PI3, SERPINB3, SERPINB4 and DSC2, as the upregulated genes and FLG, FLG2, DUSP1, XIST, CLEC2A, CDR1, KRT2, SERPINA12, GAN, MIR4720 as the downregulated genes. Insights into the role of these genes in physiology and metabolism is critical for determining the alterations or deterioration that their dysregulation may produce.

As validated from previous research the role of deregulated genes are as follows.

In the current investigation the long non-coding RNA IFITM4P was upregulated. This lncRNA phosphorylates TAK1 which is a significant member of innate immunity and pro-inflammatory signaling pathway. The deactivation of TAK1 leads to the activation of caspase 8 signaling pathways which induces psoriasis like condition [63]. The highly expressed gene is IFITM4P, IFITM4P was upregulated by interferon (IFN) signaling during viral infection as it is a positive regulator of innate antiviral immunity and altering the expression of this lncRNA had significant effects on the mRNA levels of several members of the IFITM family including IFITM1, IFITM2 and IFITM3 [64].

The p13 are required for the normal functioning of mitochondria. The pathogenetic expression of psoriasis which greatly alter based on the metabolic and signaling functions of mitochondria. Mitochondria is involved in various cellular functions like oxidative phosphorylation, regulation of the intracellular calcium levels, generation of the reactive oxygen species, apoptosis initiation etc. The p13 over expression induces mitochondrial dysfunction [65].

SERPINB3 are serine/cysteine protease inhibitors whose overexpression is associated with various pathological conditions including dermal allergies, psoriasis, and atopic dermatitis [66]. TEAD4 is an important regulatory factor in psoriasis progression, and TEAD4-mediated crosstalk

between keratinocytes and T cells plays a crucial role in the pathogenesis of psoriasis. TEAD4 can transcriptionally regulate expression of SERPINB3/4 and affect chemokine secretion, and expression of SERPINA12 and depletion of SERPINB3/4 inhibited chemokine secretion [67]. DSC2 is a calcium dependent glycoprotein that is vital for cell adhesion and cell signaling [68].

The FLG gene is required to maintain the structural integrity of the epidermal layer. The downregulation of this gene is often linked increased dermal allergies as the downregulation causes dehydration and disrupt the pH of the skin, making the skin prone to allergies and infection [69]. Filaggrin is a key protein involved in the skin's barrier function. Mutations in the filaggrin gene (FLG) have been identified [70]. It also facilitates the terminal differentiation of the epidermis and the formation of the skin barrier. Mutations in the filaggrin gene (FLG) have been identified as the cause of ichthyosis vulgaris and have been shown to be important predisposing factors for the skin disease [71].

The CLEC2A are expressed in the dermal tissues and portray a vital role in the immunological skin scrutiny [72]. The KRT2 genes determines the coat color of the dermal layer therefore, its expression is directing associated with melanogenesis [73]. The GAN genes are vital for maintaining the dermal health. These gene generate a protein termed as gigaxonin which scavenges the damaged proteins and structures within the cell [74].

10.3- Target prediction analysis

Several genes cannot be expressed but can be expressed in the RNA, and they are engaged in gene regulation through RNA interference, which successfully subdues the translation. The fact that a particular miRNA can influence the activity and expression of many genes and that multiple miRNA can regulate the functionality and expressions of a particular gene continues to be an important determinant in the advancement of various disorders. Identifying the miRNAs that are significant in psoriasis pathogenesis and their regulatory mechanisms, as well as any apparent alteration in their activity that has added to the pathology, is crucial since it provides crucial information for deeper understanding the disease.

The study revealed the presence of hub miRNAs which are localized in various dermal cells. The hsa-miR-21, hsa-miR-31, hsa-miR-203, hsa-miR-143, hsa-miR-149, hsa-miR-155, hsa-miR-146a, hsa-miR-17-3p, hsa-miR-200c are predominantly found in keratinocytes and demonstrated

wound healing properties. While hsa-miR-203, hsa-miR-7, hsa-miR-573-3p are aggregated in the meibocyte and demonstrates anti-inflammatory and protective properties. The hsa-miR-21, hsa-miR-203, hsa-miR-29b, hsa-miR-98, hsa-miR-185 are in the fibroblast cells and are critical for wound healing. Similarly, the mRNAs like hsa-miR-34a, hsa-miR-145, hsa-miR-124, hsa-miR-23a, hsa-miR-101, hsa-miR-155, hsa-miR-146a, hsa-miR-21-3p are involved in photoaging. The mRNAs hsa-miR-143, hsa-miR-155-5p, hsa-miR-214, hsa-miR-146a, hsa-miR-106b, hsa-miR-203 work as skin barriers [76] [90-94].

The following table outlines the miRNA validation approached in wet lab and the high throughput strategies

Table 7: miRNA validation methodologies

Type	**Wet lab techniques**	**High throughput techniques**
miRNA and mRNA co-expression studies	Microarray Northern botting qPCR IHC assays	RNA-seq
miRNA and mRNA specific interaction evaluation	Luciferase assay	CLIP-seq CLASH
Effect of miRNA on the protein expression pattern	Western blotting ELISA ICC assay	Psilac 2DIGE Translational profiling
miRNA biological functions	Biochemical pathway evaluation Cellular assays	Pathway enrichment analysis

The experimental methods used to validate miRNA targets are limited while there are many prediction tools and therefore, implanting these prediction tools are beneficial as they serve as pre-validation tools. High-throughput approaches have the potential to fill the gap amongst computer prediction and experimental confirmation [95]. They are effective at spotting actual miRNA-mRNA associations and evaluating miRNA-induced consequences because they have better resolution. Deep sequencing, on the other hand, not only costs time and resources, but it

also identifies numerous miRNAs whose physiological process is unknown and generates a huge data that is hard to comprehend. In any case, conventional techniques must be used to corroborate High-throughput data to discover individual miRNA-mRNA associations [96].

The disparity between transcriptomic and proteomic results is critical when interpreting miRNA datasets. Transcriptomic approaches look for mRNAs of significance but can't locate restricted target while the proteomic approaches cannot distinguish between fragmentation and suppression but do not overlook any inhibitory activity [97]. It's also worth noting that gene expression and physicochemical techniques investigate two distinct elements of miRNA activity. The first approach concentrates on regulatory potential whereas biochemical experiments are more concerned with identifying miRNA receptors and less focused on the regulatory activity [98]. The visualization tools help in better visualization and analysis of these data.

10.4- Hub gene module identification analysis

Multiple genes have been implicated in the etiology of psoriasis and some of these genes were differentially expressed in diseased skins, but the proportional contribution of each gene is yet unknown. To forecast prognosis and design alternative potential therapeutics, it is critical to discover sensitive and reliable biomarkers specific for psoriasis pathogenesis and unravel their underlying molecular pathways. Computational study of hub genes is reliable for evaluating the pathophysiology of complicated disorders [99].

According to this Hub, lncRNA and Hub, circRNA are identified; What is meant are those entries that show the highest number of interactions with different miRNAs. CytoHubba plugin was used to find the degree of nodes. A degree cutoff of greater than or equal to 7 was set for lncRNA and a degree cutoff of greater than or equal to 4 was set for circRNA. IFI44L, HK1, STAT1 and Hub lncRNA are XIST, NEAT1, KCNQ1OT1, MALAT1, HCG18, NORAD.

The relevance of these 9 hub genes as corroborated by past research is documented in Table 7

Table 8: Significance of the hub genes identified in the study

Hub gene		**Role**
circRNA	IFI44L	Antiviral and tumor suppressor [100] Innate immunity Inflammatory response [101]
	HK1	Postischemic neovascularization [102]
	STAT1	Skin hydrating role with STAT1 upregulation [103] STAT1 defect results in skin impairment [104]
lncRNA	XIST	Synthesis of extracellular matrix and aids in wound healing [104][105]
	NEAT1	Regulates the transcription of viral genome [106]
	KCNQ1OT1	Regulates keratinocyte migration Promotes wound healing [107].
	MALAT1	Promotes fibroblast migration [108] Regulates cell proliferation, invasion, apoptosis [109]
	HCG18	Promotes cell proliferation in cancer [110]
	NORAD	Regulates keratinocyte proliferation [111]

It is important to note that among the 9 hub genes, IFI44L demonstrates anti-viral potency while most of the hub genes have anti-inflammatory, migration, and infiltration properties in common. Also, these hub genes are involved in various cellular mechanisms which are required to maintain the structural integrity of the dermal layer.

10.5- PPI network analysis

The fundamental drivers of physiological functions are proteins. Undoubtedly, proteins are involved in biomolecular and cellular pathways thereby influences the health of the organism. But the proteins are not active as isolated entities; instead, they engage with one another and with other components such as DNA and RNA to execute their biological roles. As a result, investigating protein - protein interaction is critical in comprehending their function within the cell. Because these interactions can occur in various forms, the term PPI (Protein-protein interaction) represents a wide range of internal cellular activities. A PPI might suggest the

creation of steady or temporary protein complexes, physical associations, as well as physiological or biological interactions.

To discover PPI's and to generate PPI networks, a variety of approaches have been developed. In general, none of these methods is superior to the others, but every method has its own advantages and disadvantages. The sensitivity and specificity of a procedure are used to assess its effectiveness. Even though PPI networks are inherently imperfect and error-prone, they are crucial in a variety of situations. The examination of such networks provide insight into the mechanisms that cause disease onset and development. Hub proteins play a pivotal role in the preponderance of interactions, whereas most proteins that aren't hubs are only involved in a limited proportion of connections.

A protein-protein interaction (PPI) network was constructed for DEGs based on protein databases, followed by a module analysis. The STRINGdb plugin in Cytoscape was used for protein-protein interaction. The strongly interacting proteins are connected. The ceRNA network was constructed based on the interaction between miRNAs and mRNAs and lncRNAs/circRNAs and miRNAs. The hub gene module was identified from the PPI network. A total of 8 genes are found that are strongly linked CCNB1, DLGAP5, ANLN, KPNA2, STAT1, OAS2, IFI44L EIF2KA2.

Previous studies suggested that the CCNB1 gene might regulate the proliferation and apoptosis of pituitary tumor cells and activate EMT [112]. It was shown that the CCNB1 protein encoded by the CCNB1 gene belongs to the cyclin superfamily and is expressed in almost all human tissues [113]. CCNB1 can form a complex with p34, also known as cdc2, which forms the factor promoting maturation and is necessary for proper control of the G2/M transition phase of the cell cycle [114]. In normal cells, CCNB1 is expressed in late S stage, increases significantly in G2 phase and peaks in M-stage [114][115].

The genes with a high degree of protein-protein connectivity in the PPI network play a key role in the network, including DLGAP5. DLGAP5 was one of the top 8 genes with the highest level of protein-protein connectivity. DLGAP5-associated genes were mainly involved in cell cycle control, DNA replication, oocyte meiosis, p53 signaling and pyrimidine metabolism [116].

10.6- Pathway Analysis

Pathway analysis is a collection of frequently used techniques for that aims to analyze of high-throughput biological data. The strategy of these tools is based on the collection and use of biomolecular information, as well as statistical testing and other algorithms. Despite their widespread use, the underpinnings and broad context of pathway analysis may be misunderstood, leading to misrepresentation of research findings.

The most common pathway associated with psoriasis is identified as cell cycle deregulation. Other interconnected cellular mechanisms involve angiogenesis, keratinocyte proliferation, secretion pathways, homeostasis process, immune response, cell division and differentiation mechanisms, apoptotic processes, pathogenesis, innate immune response, etc [117]. The homeostasis of the skin can only be maintained when there is an equilibrium between the cell and exogenous pathogens. With an altered equilibrium the dermal cells are more prone to infections [118][119]. When the cell cycle is deregulated, there is a consequent misfunctioning of cell cycle checkpoints as a result there will be aberrant expression of incomplete proteins and chromatins which further facilitates the disease progression [120].

Recent studies show that KPNA2 expression is also induced in tumor cells and is strongly associated with prognosis. KPN2 is interlinked with the apoptotic processes, cell differentiation and innate response pathways of psoriasis. Interleukins (IL6, IL12B, IL1B, IL37, IL17A IL23A,) TNF, IFNG, and NLS are the common genes in the pathway. One of the earlier studies shows for the first time that the level of phosphorylation of STAT1 (Tyr701) and STAT1 (Ser727) is increased in psoriatic skin cells. In addition, specific signaling pathways leading to this phosphorylation have been identified and increased, and the data indicate an important role for STAT1 in the pathogenesis of psoriasis [122]. Regulation of sister chromatid segregation and regulation of cell cycle metaphase/anaphase transition were some of the significantly improved signaling pathways for the selected genes.

Conclusion

The continuous debate over the biochemical nature, and precedence complicated interplay of the immunological moieties like epidermal keratinocytes, T cells, macrophages, epithelial cells, neutrophils, sensory nerves, etc. has pushed experimental dermatology to greater heights. There's no doubt that breakthroughs in molecular immunological discoveries, human genetics and advancements in computational biology have improved knowledge of the disease's pathogenesis and management.

11- Limitations

The limitations of the current study can be addressed by conducting the research on a larger data set and by categorizing the expression pattern of the genes in accordance with disease subtype, age, gender, co-morbidities, genetic and other factors that influence the disease progression.

12- Research Ethics

I hereby declare that I have proposed the current work with my own ideas and through the initiative of gaining insights from previous work. I acknowledge that scientists have done remarkable work in this field, as their work has enabled me to create my proposal and to draw useful conclusions from my research. My work relied entirely on bioinformatics tools and software running in their default settings. In research, there are always some ethical issues in using human-derived data or samples. Participation in research must always be consented to. The data used in this study was publicly available, so ethical approval was not required. Another aspect to consider is that using bioinformatics tools other than those planned from the beginning or adding new analyzes to the data could lead to new results that could be useful for the participants to know or not to know, and then there might be a discussion of handling new information. Previous work on the associated software and web links for some of the software are duly acknowledged. All relevant literature such as research papers and publications are correctly cited in my work.

The tables and figures supporting my findings are properly documented. In my work, no figures and tables were changed, or incorrect results documented. I have tried my best to do my research in a timely and honest manner. I am grateful to my supervisors for their time to help me reconstruct and improvise my proposal, and for their constant mentoring, guidance, and feedback that has helped me evaluate my work to achieve better results. I would like to thank the university for giving us the opportunity to do research alongside the curriculum.

13- Impact on society

Psoriasis is not a life-threatening disease, but it does have social implications. The social impact of psoriasis is significant. In addition to physical appearance and pain, psoriasis imposes psychological and social burdens on sufferers (e.g., effects on interpersonal relationships, sexual function, intimacy, and professional success). Most people affected by psoriasis experience disease-related impairments in everyday activities such as work, school, interpersonal relationships, leisure activities, and intimacy. The present study aimed to comprehensively examine the variations in the expression levels of circular

RNA (circRNA), long non-coding RNA (lncRNA), microRNA (miRNA/miR) and mRNA in psoriasis. The study implemented PPI network construction methodologies to identify significant mRNAs associated with the disease's progression. The pathway analysis revealed that the cell division and differentiation are the key regulatory mechanisms responsible for disease pathogenesis. We still don't have a comprehensive solution for this ubiquitous and enigmatic ailment, and we haven't conclusively found the key genes or antigens responsible for the condition. The expression pattern of the genes is highly varied from patient to the patient making the prognosis and management more complex.

14 – Acknowledgement:

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